

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

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Approaches tried

- Tested affine and multiplicative candidates. This identified the family $f(x) = x + c$ with $c \geq 0$; direct substitution turns both squared inequality gaps into $(x - y - c)^2$.
- Used the equality-producing substitution $x = f(y)$. It gives the exact second-iterate relation $f(f(y)) = 2f(y) - y$.
- Iterated the second-iterate relation. Every forward orbit is the arithmetic progression $f^n(x) = x + n(f(x) - x)$, so positivity of the codomain forces $f(x) \geq x$.
- Rewrote the upper inequality in terms of the displacement $d(x) = f(x) - x$. For two strictly positive displacements, synchronized their arithmetic orbits to remain a bounded distance apart while tending to infinity. Applying the upper inequality in both orders forces the two displacements to be equal.
- Ruled out coexistence of zero displacement and a common positive displacement. The upper inequality separates points of the two types by more than the positive displacement, while a finite chain in an interval with shorter steps must contain two adjacent points of different types.

- Also examined pointwise bounds obtained from $x = y$ and the lower inequality; these give rough growth restrictions but do not by themselves compare displacements at different points, so they were not used in the final proof.

Current best

All solutions are exactly $\boxed{f(x) = x + c \quad (x > 0)}$, where c is an arbitrary real constant with $c \geq 0$.

Full proof

Define the displacement

$$d(t) := f(t) - t \quad (t > 0).$$

We first obtain an exact relation. In the given chain put $x = f(y)$. Its leftmost term is then

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y),$$

and its rightmost term is

$$\sqrt{f(y)f(y)} = f(y).$$

Consequently the middle term is equal to $f(y)$, and hence

$$f(f(y)) = 2f(y) - y. \quad (1)$$

It follows that

$$d(f(y)) = f(f(y)) - f(y) = f(y) - y = d(y). \quad (2)$$

By induction, (2) gives, for every integer $n \geq 0$,

$$f^n(y) = y + nd(y). \quad (3)$$

If $d(y) < 0$, then the right-hand side of (3) is nonpositive for every sufficiently large n , contrary to the fact that every iterate of f lies in $\mathbb{R}_{>0}$. Thus

$$d(y) \geq 0 \quad \text{for every } y > 0. \quad (4)$$

We next prove that any two positive values of d are equal. Squaring the first inequality in the given chain is legitimate because both sides are positive. On writing

$$a = d(x), \quad b = d(y),$$

we obtain

$$2x^2 + 2(y + b)^2 \geq (x + a + y)^2,$$

or equivalently

$$(x - y)^2 + 4yb + 2b^2 \geq 2a(x + y) + a^2. \quad (5)$$

Let $u, v > 0$ satisfy

$$d(u) = a > 0, \quad d(v) = b > 0.$$

By (2)–(3),

$$d(u + ma) = a, \quad d(v + nb) = b \quad (6)$$

for all integers $m, n \geq 0$. For every sufficiently large integer m , put

$$X_m = u + ma, \quad n_m = \left\lfloor \frac{X_m - v}{b} \right\rfloor, \quad Y_m = v + n_m b.$$

Then $n_m \geq 0$ and

$$0 \leq X_m - Y_m < b. \quad (7)$$

In particular, $X_m, Y_m \rightarrow \infty$ and $X_m/Y_m \rightarrow 1$. By (6), their displacements are respectively a, b . Apply (5) with $(x, y) = (X_m, Y_m)$ and divide by Y_m :

$$\frac{(X_m - Y_m)^2}{Y_m} + 4b + \frac{2b^2}{Y_m} \geq 2a \left(\frac{X_m}{Y_m} + 1 \right) + \frac{a^2}{Y_m}.$$

Letting $m \rightarrow \infty$ and using (7) yields $4b \geq 4a$, so $b \geq a$. Applying (5) instead with $(x, y) = (Y_m, X_m)$ and dividing by X_m gives, in the same way, $4a \geq 4b$. Therefore $a = b$.

It follows that either d is identically zero, or there is a number $c > 0$ such that

$$d(t) \in \{0, c\} \quad (t > 0). \quad (8)$$

We show that in the latter case the value 0 cannot occur. Suppose that $d(p) = c$ and $d(q) = 0$. Substitution of $x = p$, $y = q$, $a = c$, and $b = 0$ into (5) gives

$$(p - q)^2 \geq 2c(p + q) + c^2 > c^2,$$

so every point with displacement c is at distance greater than c from every point with displacement 0:

$$|p - q| > c. \quad (9)$$

If both types in (8) occurred, choose one point p of the first type and one point q of the second type. Choose an integer $N > |p - q|/c$ and divide the interval from p to q into the points

$$t_i = p + \frac{i}{N}(q - p), \quad i = 0, 1, \dots, N.$$

All these points are positive, and consecutive ones are at distance $|p - q|/N < c$. Since t_0 and t_N have different displacement types, some consecutive pair must have different types. This contradicts (9). Hence no mixing is possible. Together with (4), we have proved that for some constant $c \geq 0$,

$$d(t) = c \quad (t > 0),$$

and therefore

$$f(t) = t + c. \quad (10)$$

It remains to verify these functions. Let $c \geq 0$ and $f(t) = t + c$. For every $x, y > 0$, the arithmetic mean of the two positive numbers x and $y + c$ is

$$\frac{x + y + c}{2} = \frac{f(x) + y}{2}.$$

The root-mean-square–arithmetic-mean and arithmetic-mean–geometric-mean inequalities give

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2} \geq \sqrt{x(y + c)},$$

which is exactly the required chain. Thus all and only the functions in (10), with $c \geq 0$, are solutions.